

Trigonometry

Week 1A

Overview

Trigonometry

Key Ideas - Right-Angle Triangles

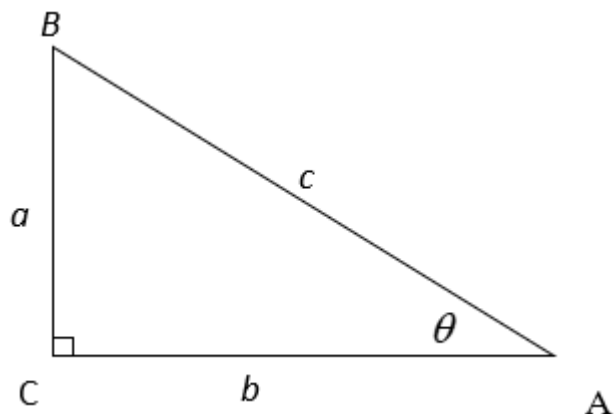
- Trigonometric Ratios*
- Trigonometric Ratios of Complementary Angles*
- Trigonometric Ratios on the Calculator*
- Right-angled Triangle Problems*
- Exact Trigonometric Ratios*

Key Idea – Right-angled Triangles

Trigonometry is a branch of Mathematics that focuses on the relationship between the measurement of sides and angles in triangles. Trigonometry is used to solve many practical problems and to understand natural phenomena.

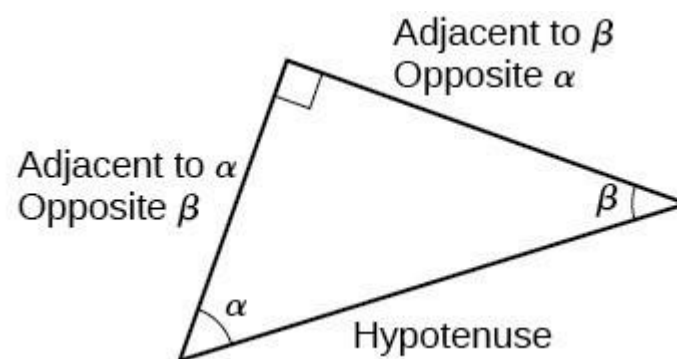
In a right-angled triangle (a triangle where one of the angles is 90°):

- The **hypotenuse** is the longest side (side c) and is always opposite the right-angle.
- Side a is **opposite** the reference angle θ .
- Side b is **adjacent** to the reference angle θ .



***Note:** The vertices (corners) of a triangle are labelled with capital letters and the opposite sides with the corresponding small case letters.*

The positions of the opposite and adjacent sides vary according to where the angle is marked.

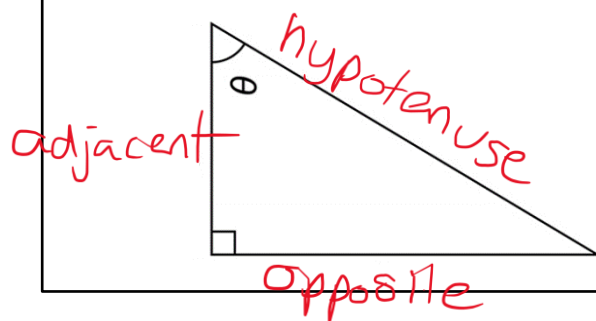


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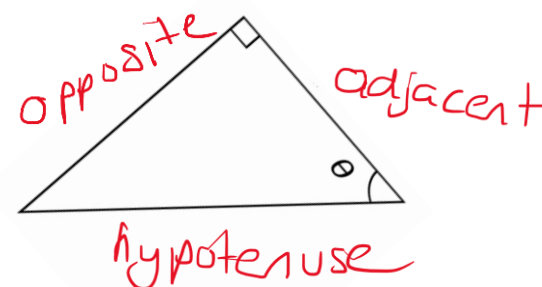
Exercise:

For each of the following triangles, label each side as opposite, adjacent or hypotenuse.

(a)



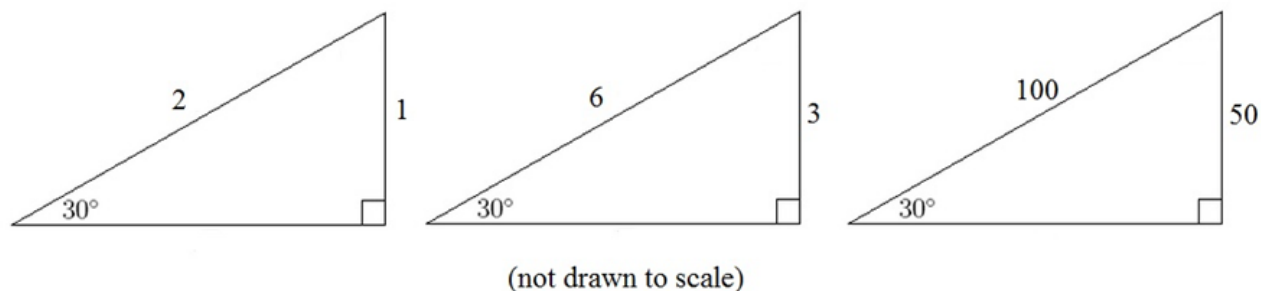
(b)



Key Idea – Trigonometric Ratios

The study of Trigonometry is based on the observation that, in similar right-angled triangles (those with another matching angle), the ratios of the sides of the triangles are constant.

Example:



$$\frac{\text{opposite}}{\text{hypotenuse}} = \frac{1}{2}$$

$$\frac{\text{opposite}}{\text{hypotenuse}} = \frac{3}{6} \\ = \frac{1}{2}$$

$$\frac{\text{opposite}}{\text{hypotenuse}} = \frac{50}{100} \\ = \frac{1}{2}$$

The ratios $\frac{\text{opposite}}{\text{adjacent}}$, $\frac{\text{adjacent}}{\text{hypotenuse}}$, $\frac{\text{opposite}}{\text{hypotenuse}}$ are constant for any given value of θ .

The ratios of the specific sides in a right-angled triangle are called sine (sin), cosine (cos) and tangent (tan) and are defined as:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{o}{h}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{a}{h}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{o}{a}$$

It is important to remember the trigonometric ratios for sine, cosine and tangent!

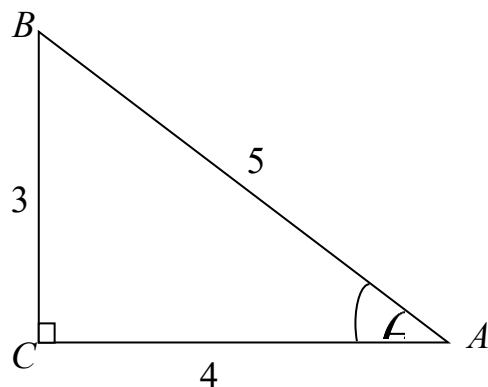
A common way to recall these ratios is by remembering:

Soh Cah Toa

***Note:** Since the hypotenuse is the longest side in a right-angled triangle, the values of $\sin \theta$ and $\cos \theta$ cannot be greater than 1.*

Example:

In the right-angled triangle ABC , $a = 3$, $b = 4$, and $c = 5$. Use this information to give the trigonometric ratios.



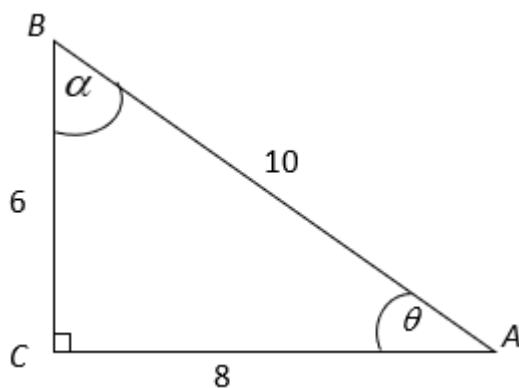
$$\sin \theta = \frac{o}{h} = \frac{3}{5}$$

$$\cos \theta = \frac{a}{h} = \frac{4}{5}$$

$$\tan \theta = \frac{o}{a} = \frac{3}{4}$$

Exercise:

In the right-angled triangle, ABC



Find

$$(a) \sin \alpha = \frac{o}{h} = \frac{8}{10} = \frac{4}{5}$$

$$(b) \cos \theta = \frac{a}{h} = \frac{8}{10} = \frac{4}{5}$$

$$(c) \tan \theta = \frac{o}{a} = \frac{6}{8} = \frac{3}{4}$$

Reciprocal Trigonometric Ratios

The reciprocal ratios cotangent (cot), cosecant (cosec) and secant (sec) are defined as:

$$\text{cosec } \theta = \frac{1}{\sin \theta}, \quad \sin \theta \neq 0$$

$$\sec \theta = \frac{1}{\cos \theta}, \quad \cos \theta \neq 0$$

$$\cot \theta = \frac{1}{\tan \theta}, \quad \tan \theta \neq 0$$

It is important to remember these relationships!

$$\text{cosec } \theta = \frac{\text{hypotenuse}}{\text{opposite}}, \quad \sin \theta \neq 0$$

$$\sec \theta = \frac{\text{hypotenuse}}{\text{adjacent}}, \quad \cos \theta \neq 0$$

$$\cot \theta = \frac{\text{adjacent}}{\text{opposite}}, \quad \tan \theta \neq 0$$

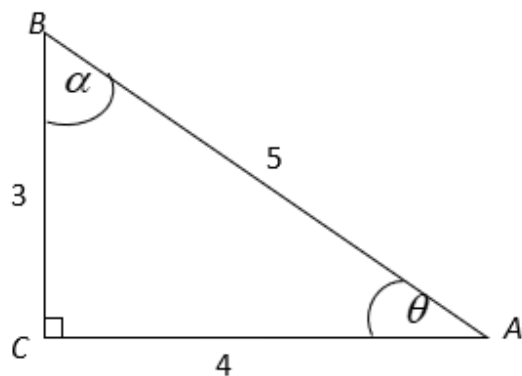
The reciprocal ratios can be derived from the ratios for sine, cosine and tangent.

Also $\cot \theta = \frac{\cos \theta}{\sin \theta}$

Note: The abbreviation $\csc \theta$ is also used for the cosecant θ .

Example:

In the right-angled triangle ABC , $a = 3$, $b = 4$, and $c = 5$. Find the value of $\sin \theta$, $\sec \theta$ and $\cot \alpha$.



Recall:

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

(a) Find $\operatorname{cosec} \theta$

$$\begin{aligned}\sin \theta &= \frac{o}{h} \\ &= \frac{3}{5}\end{aligned}$$

$$\operatorname{cosec} \theta = \frac{5}{3}$$

(b) Find $\sec \theta$

$$\begin{aligned}\cos \theta &= \frac{a}{h} \\ &= \frac{4}{5}\end{aligned}$$

$$\sec \theta = \frac{5}{4}$$

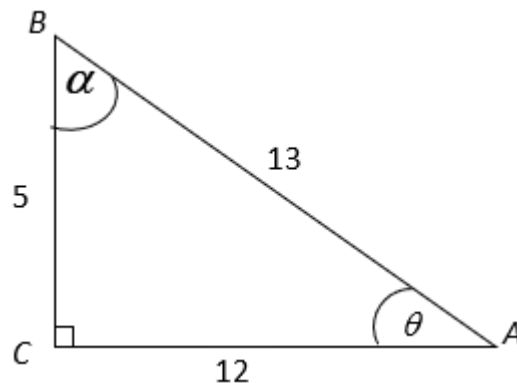
(c) Find $\cot \alpha$

$$\begin{aligned}\tan \alpha &= \frac{o}{a} \\ &= \frac{4}{3}\end{aligned}$$

$$\cot \alpha = \frac{3}{4}$$

Exercise:

In the right-angled triangle ABC



Find the value of:

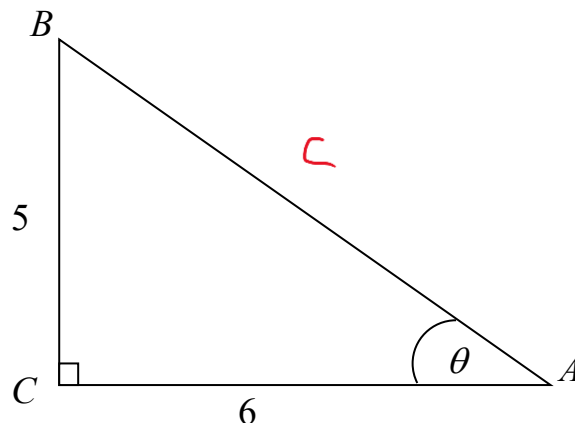
$$\begin{aligned} \text{(a) } \operatorname{cosec} \alpha &= \frac{1}{\sin \alpha} \\ &= \frac{1}{\frac{5}{13}} \\ &= \frac{13}{5} \end{aligned}$$

$$\begin{aligned} \text{(b) } \sec \theta &= \frac{1}{\cos \theta} \\ &= \frac{1}{\frac{12}{13}} \\ &= \frac{13}{12} \end{aligned}$$

$$\begin{aligned} \text{(c) } \cot \theta &= \frac{1}{\tan \theta} \\ &= \frac{1}{\frac{5}{12}} \\ &= \frac{12}{5} \end{aligned}$$

Discussion Questions:

Consider triangle ABC below.



- (a) What additional information is needed to find the value of $\sin \theta$ and $\cos \theta$?

$$\sin \theta = \frac{5}{c} \quad \cos \theta = \frac{6}{c} \quad \text{we need } c$$

- (b) How could this missing information be calculated?

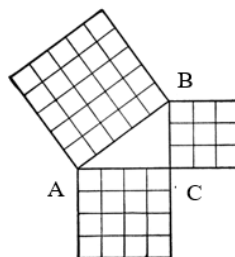
Pythagoras' theorem

In questions where we are not given the length of the third side, we need to use Pythagoras' Theorem. In these cases, you need to draw a diagram and label it carefully.

Pythagoras' Theorem:

$$(AB)^2 = (BC)^2 + (AC)^2$$

$$c^2 = a^2 + b^2$$



"The square of the hypotenuse is equal to the sum of the squares of the other two sides."

Example:

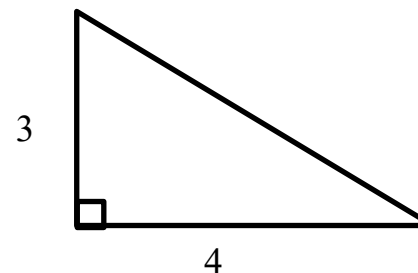
Use Pythagoras theorem to find the length of the hypotenuse.

$$c^2 = 3^2 + 4^2$$

$$c = \sqrt{3^2 + 4^2}$$

$$c = \sqrt{25}$$

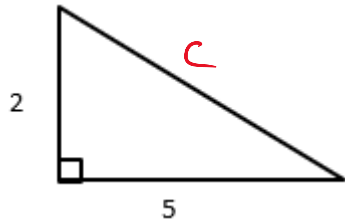
$$= 5$$



Note: It is very useful to remember the right-angled triangle with side lengths 3, 4 and 5.

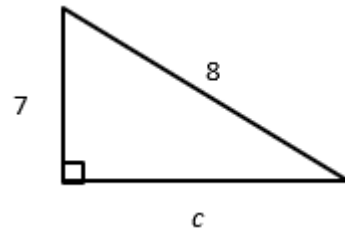
Exercise:

- (a) Use Pythagoras theorem to find the length of the hypotenuse in **exact form**.



$$\begin{aligned}c^2 &= 2^2 + 5^2 \\&= 29 \\c &= \sqrt{29}\end{aligned}$$

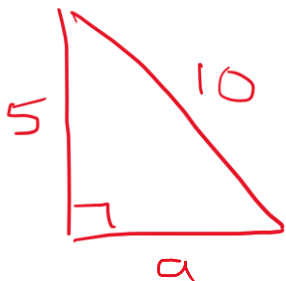
- (b) Use Pythagoras theorem to find the exact length of the side c.



$$\begin{aligned}8^2 &= 7^2 + c^2 \\c^2 + 7^2 &= 8^2 \\c^2 &= 8^2 - 7^2\end{aligned}$$

$$\begin{aligned}c &= \sqrt{8^2 - 7^2} \\&= \sqrt{15}\end{aligned}$$

- (c) In a right-angled triangle, the length of one side is 5cm and the length of the hypotenuse is 10cm. Find the length of the third side correct to two decimal places.



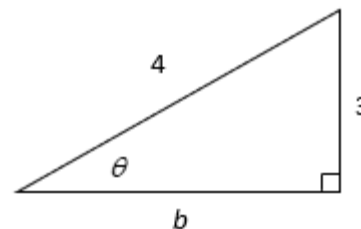
$$\begin{aligned}a^2 + 5^2 &= 10^2 \\a^2 &= 10^2 - 5^2 \\&= 75\end{aligned}$$

$$\begin{aligned}a &= \sqrt{75} \\&= \sqrt{25} \sqrt{3} \\&= 5\sqrt{3}\end{aligned}$$

$$= 8.66 \text{ cm}$$

Example: If angle θ is found in a right-angle triangle with hypotenuse of length 4 units and the side opposite θ has length 3 units, find the **exact value** of $\tan \theta$.

Step 1. Draw and label triangle



Step 2. Use Pythagoras' Theorem

$$4^2 = 3^2 + b^2$$

$$b^2 = 4^2 - 3^2$$

$$b = \sqrt{7}$$

Step 3. Find $\tan \theta$

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{3}{\sqrt{7}} = \frac{3\sqrt{7}}{7}$$

Exercise: In a right-angle triangle, $\sin \theta = \frac{2}{3}$. Find the exact value of $\cos \theta$.



$$a^2 + 2^2 = 3^2$$

$$a^2 = 3^2 - 2^2$$

$$= 5$$

$$a = \sqrt{5}$$

$$\begin{aligned} \cos \theta &= \frac{a}{3} \\ &= \frac{\sqrt{5}}{3} \end{aligned}$$

Exercise:

If angle θ is found in a right-angle triangle and:

(a) $\tan \theta = \frac{1}{4}$ find exact value of $\sec \theta$



$$c^2 = 1^2 + 4^2$$

$$= 17$$

$$c = \sqrt{17}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$= \frac{1}{\frac{4}{c}}$$

$$= \frac{c}{4}$$

$$= \frac{\sqrt{17}}{4}$$

$$= \frac{\sqrt{17}}{4}$$

(b) $\sec \theta = \frac{17}{15}$ find exact value of $\cos \theta$.



$$a^2 + 15^2 = 17^2$$

$$a^2 = 17^2 - 15^2$$

$$= 64$$

$$a = \sqrt{64}$$

$$= 8$$

$$\cos \theta = \frac{a}{h}$$

$$= \frac{8}{17}$$

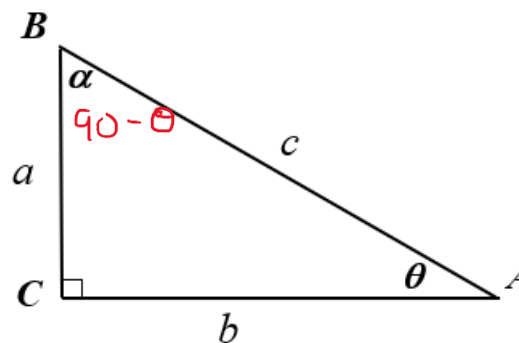
Key Idea – Trigonometric Ratios of Complementary Angles

A right-angled triangle will have two complementary angles.

The sum of complementary angles is 90° . $\theta^\circ + \alpha^\circ = 90^\circ$

Discussion Questions:

Find the following ratios and consider the relationships between the ratios of complementary angles.



(a) $\sin \theta = \frac{a}{c}$

(e) $\cos(90^\circ - \theta) = \frac{a}{c}$

(b) $\cos \theta = \frac{b}{c}$

(f) $\sin(90^\circ - \theta) = \frac{b}{c}$

(c) $\tan \theta = \frac{a}{b}$

(g) $\cot(90^\circ - \theta) = \frac{a}{b}$

(d) $\operatorname{cosec} \theta = \frac{c}{a}$

(h) $\sec(90^\circ - \theta) = \frac{c}{a}$

Your investigation will show that, for acute angles:

$$\sin \theta = \cos(90^\circ - \theta)$$

$$\cos \theta = \sin(90^\circ - \theta)$$

$$\tan \theta = \cot(90^\circ - \theta)$$

$$\cot \theta = \tan(90^\circ - \theta)$$

$$\sec \theta = \operatorname{cosec}(90^\circ - \theta)$$

$$\operatorname{cosec} \theta = \sec(90^\circ - \theta)$$

For example,

$$\sin 20^\circ = \cos(90 - 20)^\circ = \cos 70^\circ$$

$$\cos 5^\circ = \sin(90 - 5)^\circ = \sin 85^\circ$$

$$\tan 80^\circ = \cot(90 - 80)^\circ = \cot 10^\circ$$

Example: Simplify $\tan 50^\circ - \cot 40^\circ$.

We can see that, as 50° and 40° are complementary angles, we can use the relationship $\tan \theta = \cot(90^\circ - \theta)$.

$$\cot 40^\circ = \tan(90^\circ - 40^\circ) = \tan 50^\circ$$

$$\begin{aligned}\therefore \tan 50^\circ - \cot 40^\circ &= \tan 50^\circ - \tan 50^\circ \\ &= 0\end{aligned}$$

Exercise: Find the value of x if:

(a) $\cos x^\circ = \sin 20^\circ$

$$= \cos(90 - 20)^\circ$$

$$\cos x^\circ = \cos 70^\circ$$

$$x^\circ = 70^\circ$$

(b) $\tan 20^\circ = \cot(x + 30)^\circ$

$$\cot(90 - 20)^\circ = \cot(x + 30)^\circ$$

$$\cot(70)^\circ = \cot(x + 30)^\circ$$

$$70^\circ = x + 30^\circ \longrightarrow x^\circ = 40^\circ$$

Exercise: Without using a calculator, simplify each of the following expressions:

(a) $\tan 70^\circ + \cot 20^\circ - 2\tan 70^\circ$

$$= \tan 70^\circ + \tan(90 - 20)^\circ - 2\tan 70^\circ$$

$$= \tan 70^\circ + \tan 70^\circ - 2\tan 70^\circ$$

$$= 0$$

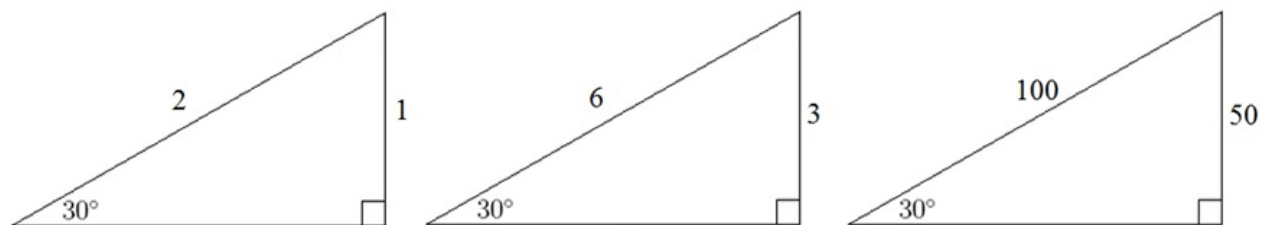
(b) $\frac{\operatorname{cosec} 55^\circ}{\sec 35^\circ} = \frac{\sec(90 - 55)^\circ}{\sec(35)^\circ}$

$$= \frac{\sec(35)^\circ}{\sec(35)^\circ}$$

$$= 1$$

Key Idea – Trigonometric Ratios on the Calculator

Recall that the ratios $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$, $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$, $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$ are constant for any given value of θ .



(not drawn to scale)

$$\frac{\text{opposite}}{\text{hypotenuse}} = \frac{1}{2}$$

$$\frac{\text{opposite}}{\text{hypotenuse}} = \frac{3}{6} \\ = \frac{1}{2}$$

$$\frac{\text{opposite}}{\text{hypotenuse}} = \frac{50}{100} \\ = \frac{1}{2}$$

Therefore, if the angle of a right angle-triangle is known then the calculator can be used to calculate the trigonometric ratios.

When performing trigonometric calculations on your calculator, it is often necessary to write your answers correct to a given number of decimal places.

Example: Use your calculator to find the following trigonometric ratio values. Give each answer correct to 4 decimal places:

$$\sin 35^\circ = 0.5736$$

$$\tan 63^\circ 46' = 2.0293$$

$$\begin{aligned}\cot 35^\circ &= \frac{1}{\tan 35^\circ} \\ &= 1.4281\end{aligned}$$

$$\begin{aligned}\sec 63^\circ 46' &= \frac{1}{\cos 63^\circ 46'} \\ &= 2.2623\end{aligned}$$

Notes: There is no cosec, sec or cot button on your calculator. These ratios are found by taking the reciprocal of the sin, cos, or tan ratio. Make sure your calculator is set in degree mode.

Exercise:

Evaluate, correct to 2 decimal places:

$$(a) \tan 35^\circ = 0.70$$

$$(b) \cos 63^\circ 46' = 0.44$$

$$\begin{aligned}(c) \operatorname{cosec} 35^\circ &= \frac{1}{\sin 35^\circ} \\ &= 1.74\end{aligned}$$

$$\begin{aligned}(d) \sec 63^\circ 46' &= \frac{1}{\cos (63^\circ 46')} \\ &= 2.26\end{aligned}$$

Converting Angle Measurements

Conversely, if the ratio is known, then the calculator can be used to determine the fixed angle in the right-angled triangle. Before going through this process, it is important to know some important information about angle measurements.

Angle measurements are usually written in degrees and minutes, where:

$$1 \text{ degree} = 60 \text{ minutes } (1^\circ = 60')$$

$$1 \text{ minute} = 60 \text{ seconds } (1' = 60'').$$

Example: $36^\circ 25' 40''$ means 36 degrees

25 minutes (meaning $\frac{25}{60}$ degree)

40 seconds (meaning $\frac{40}{60}$ minutes or $\frac{40}{3600}$ degree)

Your calculator uses degrees, minutes and seconds, so you may need to round off your answer. Angles are rounded up to the nearest degree, if there are 30 or more minutes, and up to the nearest minute, if there are 30 or more seconds.

Although we often wish to write the angle in degrees and minutes, the calculator may give the angle in decimal form. For example, the calculator may give an angle as 48.5° . This is the same as $48^\circ 30'$.

The $\boxed{\circ ' ''}$ key on a calculator changes decimal angles into degree, minutes and seconds and vice versa. Check this out, by entering 48.5 into your calculator and then pressing the $\boxed{\circ ' ''}$ key

Examples:

Write $43^\circ 10' 24''$ to the nearest minute. $\rightarrow 43^\circ 10' 24'' = 43^\circ 10'$

Write $24^\circ 33' 27''$ to the nearest degree. $\rightarrow 24^\circ 33' 27'' = 25^\circ$

Change $13^\circ 45'$ into a decimal. $\rightarrow 13^\circ 45' = 13.75$

Press 13 $\boxed{\circ ' ''}$ 45 $\boxed{\circ ' ''}$ $\boxed{=}$ $\boxed{\circ ' ''}$

Change 45.23 into degrees and minutes. $\rightarrow 45.23 = 45^\circ 13' 48'' = 45^\circ 14'$

Press 45.23 $\boxed{=}$ $\boxed{\circ ' ''}$

Exercises: Without using a calculator:

(a) Write $17^{\circ}30'29''$ (i) correct to the nearest degree, (ii) correct to the nearest minute.

18°

$17^{\circ}30'$

(b) Write the angle $20^{\circ}45'$ in degrees as a decimal.

20.75°

(c) Write the angle 40.25° in degrees and minutes.

$40^{\circ}15'$

Exercises:

(a) Use your calculator to write $91^{\circ}37'30''$ in degrees, as a decimal.

91.625°

(b) Use your calculator to write 105.02° in degrees and minutes.

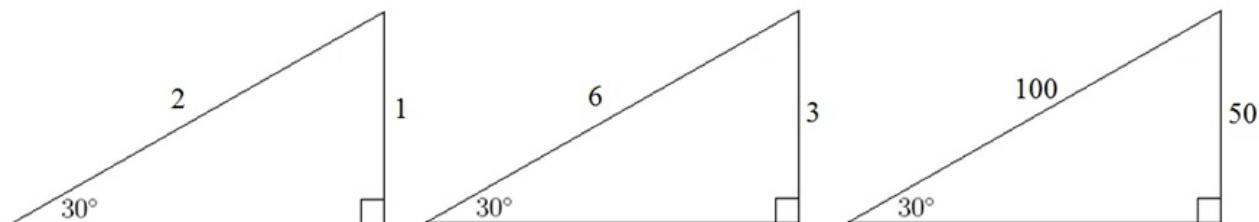
$105^{\circ}01'12''$

Finding the Size of an Angle

If a trigonometric ratio is known, then a calculator can be used to determine the size of a fixed angle in a right-angled triangle. To find the angle, we use inverse operations to make the angle the subject of the formula.

We have seen that $\sin 30^\circ = \frac{1}{2}$ so the inverse sine (arcsine) of $\frac{1}{2}$ is 30° . This is written as

$$\sin^{-1}\left(\frac{1}{2}\right) = 30^\circ.$$



(not drawn to scale)

If: $\sin \theta = x$ then $\theta = \sin^{-1}x$ or $\theta = \arcsin x$
 $\cos \theta = y$ then $\theta = \cos^{-1}y$ or $\theta = \arccos y$
 $\tan \theta = z$ then $\theta = \tan^{-1}z$ or $\theta = \arctan z$

IMPORTANT: The -1 in the index does not mean reciprocal or 'one over'. i.e. $\sin^{-1} \theta \neq \frac{1}{\sin \theta}$

Examples:

Find the size of acute angle θ *correct* to the nearest minute given $\sin \theta = 0.3874$.

$$\begin{aligned}\theta &= \sin^{-1}(0.3874) \\ &= 22.7928^\circ \\ &= 22^\circ 48'\end{aligned}$$

Find the size of acute angle θ given $\cot \theta = \frac{1}{\sqrt{3}}$

Remember $\cot \theta = \frac{1}{\tan \theta}$

$$\cot \theta = \frac{1}{\sqrt{3}}$$

$$\frac{1}{\tan \theta} = \frac{1}{\sqrt{3}}$$

$$\tan \theta = \sqrt{3}$$

$$\begin{aligned}\theta &= \tan^{-1}(\sqrt{3}) \\ &= 60^\circ\end{aligned}$$

Note: Make sure your calculator is set in **Degrees** mode.

Exercise:

Find the size of acute angle θ given:

(a) $\cos \theta = \frac{1}{\sqrt{2}}$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) \\ = 45^\circ$$

(c) $\tan \theta = 2.3874$

$$\theta = \tan^{-1}(2.3874) \\ = 67^\circ 16'$$

(b) $\sec \theta = \sqrt{2}$

$$\frac{1}{\cos \theta} = \sqrt{2}$$

$$\cos \theta = \frac{1}{\sqrt{2}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

(d) $3 \cos \theta = 1$

$$\cos \theta = \frac{1}{3}$$

$$\theta = \cos^{-1}\left(\frac{1}{3}\right) \\ = 70^\circ 32'$$

For (c) and (d) give the angle correct to the nearest minute.

Key Idea – Right-angled Triangle Problems

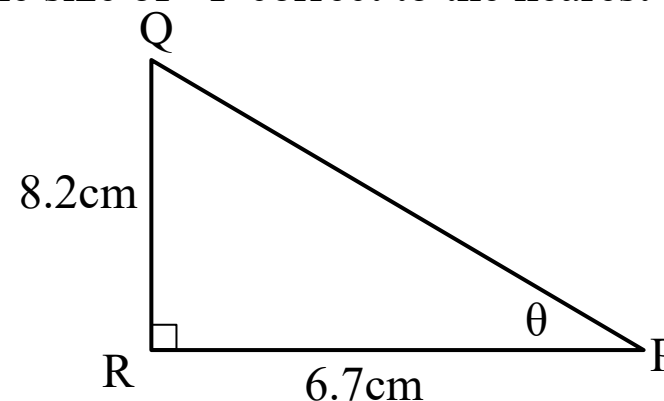
We can use trigonometry to find the size of an angle or length of a side in a right-angled triangle.

Solving Problems to find an Angle

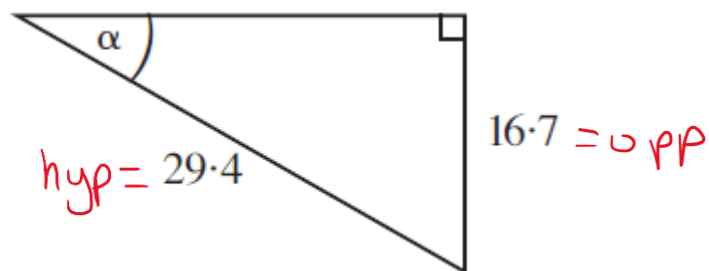
Example:

In $\triangle PQR$, $\angle R = 90^\circ$, $QR = 8.2$ cm and $PR = 6.7$ cm. Find the size of $\angle P$ correct to the nearest degree.

$$\begin{aligned}\tan \theta &= \frac{o}{a} = \frac{8.2}{6.7} \\ \theta &= \tan^{-1}\left(\frac{8.2}{6.7}\right) \\ &= 50^\circ 45' \\ &= 51^\circ\end{aligned}$$



Exercise: Find the size of the angle marked, correct to the nearest degree.

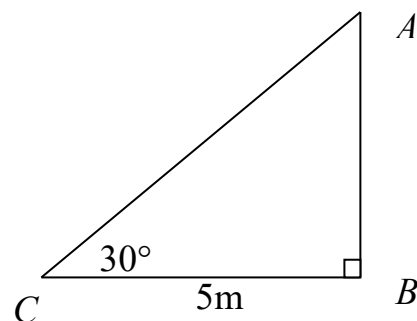


$$\begin{aligned}\sin \alpha &= \frac{16.7}{29.4} \\ \alpha &= \sin^{-1}\left(\frac{16.7}{29.4}\right) \\ &= 35^\circ\end{aligned}$$

Solving Problems to find a Side Length

Example:

Determine the lengths of sides b and c in triangle ABC , given $\angle C = 30^\circ$ and $a = 5$ metres. Give each answer in metres correct to 2 decimal places.



$$\cos \theta = \frac{a}{h}$$

$$\cos 30^\circ = \frac{5}{b}$$

$$b \times \cos 30^\circ = 5$$

$$b = \frac{5}{\cos 30^\circ}$$
$$= 5.77m$$

$$\tan 30^\circ = \frac{c}{5}$$

$$c = 5 \times \tan 30^\circ$$

$$= 2.8867....$$

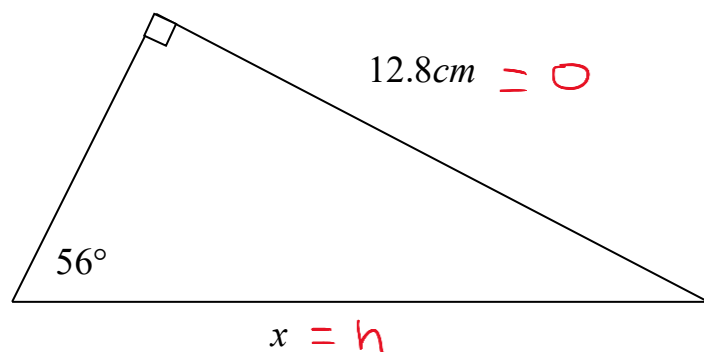
$$= 2.89m$$

Note: In the above example, the variable was in the denominator of the ratio.

Note: In the above example, the variable was in the numerator of the ratio.

Exercise:

(a) Find the length of the hypotenuse.



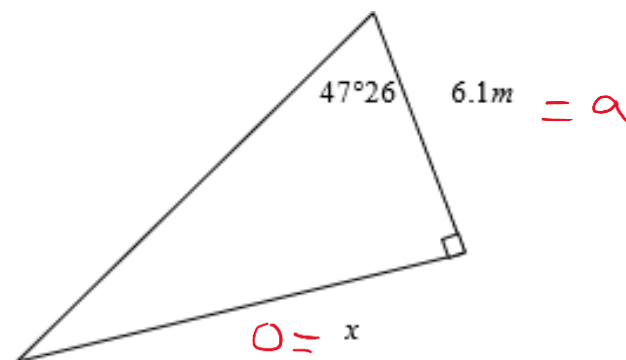
$$\sin \theta = \frac{o}{h}$$

$$\sin 56^\circ = \frac{12.8}{x}$$

$$x \sin 56^\circ = 12.8$$

$$x = \frac{12.8}{\sin 56^\circ}$$
$$= 15.44\text{ cm}$$

(b) Find the value of x , to 2 decimal places.



$$\tan \theta = \frac{o}{a}$$

$$\tan 47^\circ 26' = \frac{x}{6.1}$$

$$x = 6.1 \times \tan 47^\circ 26'$$
$$= 6.64\text{ m}$$

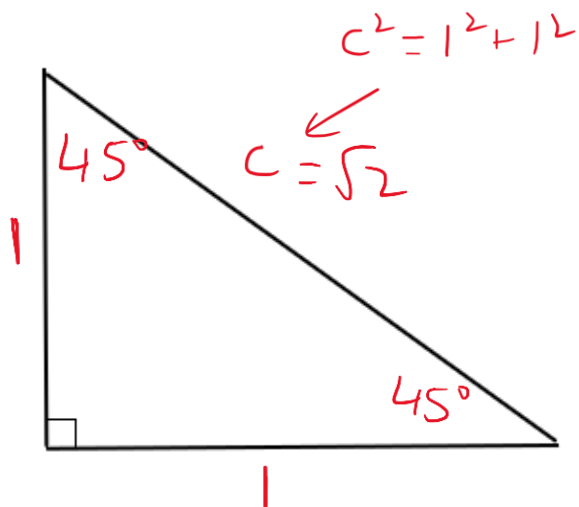
Key Idea – Exact trigonometric ratios

There are three “standard” acute angles for which you are expected to know the **exact values** of the trigonometric ratios. These angles are 30° , 45° and 60° . To find these ratios we need to remember two triangles.

45° Exact Value Ratios – Isosceles Triangle

Discussion Question:

Below is an isosceles triangle. Label the size of each angle. If the lengths of the two equal sides is 1 unit, find the length of the third side. Then determine the trigonometric ratios:



$$\sin 45^\circ = \frac{o}{h} = \frac{1}{\sqrt{2}}$$

$$\operatorname{cosec} 45^\circ = \frac{h}{o} = \frac{\sqrt{2}}{1} = \sqrt{2}$$

$$\cos 45^\circ = \frac{a}{h} = \frac{1}{\sqrt{2}}$$

$$\sec 45^\circ = \frac{h}{a} = \frac{\sqrt{2}}{1} = \sqrt{2}$$

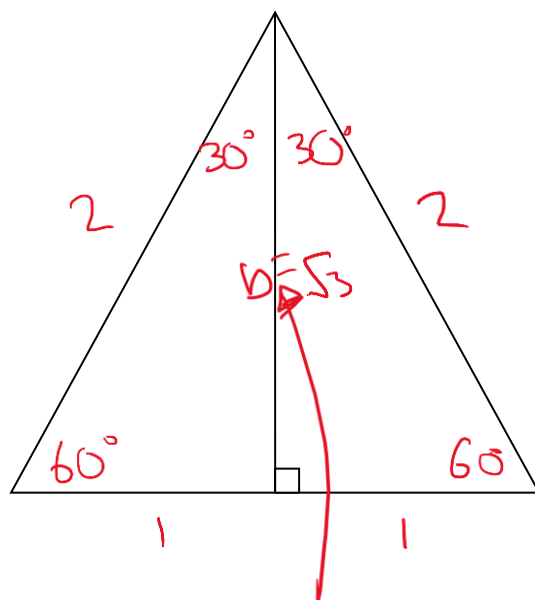
$$\tan 45^\circ = \frac{o}{a} = \frac{1}{1} = 1$$

$$\cot 45^\circ = \frac{a}{o} = \frac{1}{1} = 1$$

30° and 60° Exact Value Ratios – Equilateral Triangle

Discussion Question:

Below is an equilateral triangle with side length of 2 units. The perpendicular bisector divides the equilateral triangle into two identical triangles. Label the size of all angles in the diagram. Find the height of the equilateral triangle then determine the trigonometric ratios below:



$$\begin{aligned}b^2 + 1^2 &= 2^2 \\b^2 &= 2^2 - 1^2 \\b^2 &= 3\end{aligned}$$

$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\tan 60^\circ = \sqrt{3}$$

Summary of Exact Trigonometric Ratios of 30° , 45° , 60°

degrees	30°	45°	60°
sin	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
tan	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

Exercise:

Find the exact value of:

$$\begin{aligned}
 \text{(a) } \operatorname{cosec} 45^\circ &= \frac{1}{\sin 45^\circ} \\
 &= \frac{1}{\frac{1}{\sqrt{2}}} \\
 &= \sqrt{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) } \sec 30^\circ &= \frac{1}{\cos 30^\circ} \\
 &= \frac{1}{\frac{\sqrt{3}}{2}} \\
 &= \frac{2}{\sqrt{3}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(c) } \cot 60^\circ &= \frac{1}{\tan 60^\circ} \\
 &= \frac{1}{\sqrt{3}}
 \end{aligned}$$

Exercise:

Find the following values, giving each answer in exact form.

$$(a) \quad \cos 60^\circ \sin 45^\circ = \frac{1}{2} \times \frac{1}{\sqrt{2}} = \frac{1}{2\sqrt{2}}$$

$$(b) \quad \cos 45^\circ + \sin 45^\circ = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{1+1}{\sqrt{2}} = \frac{2}{\sqrt{2}}$$

$$(c) \quad \sqrt{\sin 30^\circ} = \sqrt{\frac{1}{2}} = \frac{\sqrt{1}}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$